GLOBAL CO_2 EMISSIONS

Global CO_2 Emissions: Trends, Regional Shifts, and Per Capita Inequalities

Scope: 1750–2024 emissions trends, regional responsibility, and per capita disparities

35+ Bt
annual global fossil CO_2 today

429 ppm
atmospheric CO_2, Feb 2026

150×
per capita emissions gap

Data sources: NASA; Our World in Data / Global Carbon Budget 2025

ATMOSPHERIC MILESTONE

Atmospheric CO_2 Reaches 429 ppm — 50% Above Pre-Industrial Levels

429 ppm

NOAA Mauna Loa Observatory measurement, Fe

~280 ppm

approximate pre-industrial baseline

>50%

increase above pre-industrial levels

Acceleration context

Pre-industrial

~280 ppm baseline

1958

modern Mauna Loa record begins

Recent centuries

rapid rise vs. millennia-scale natural changes

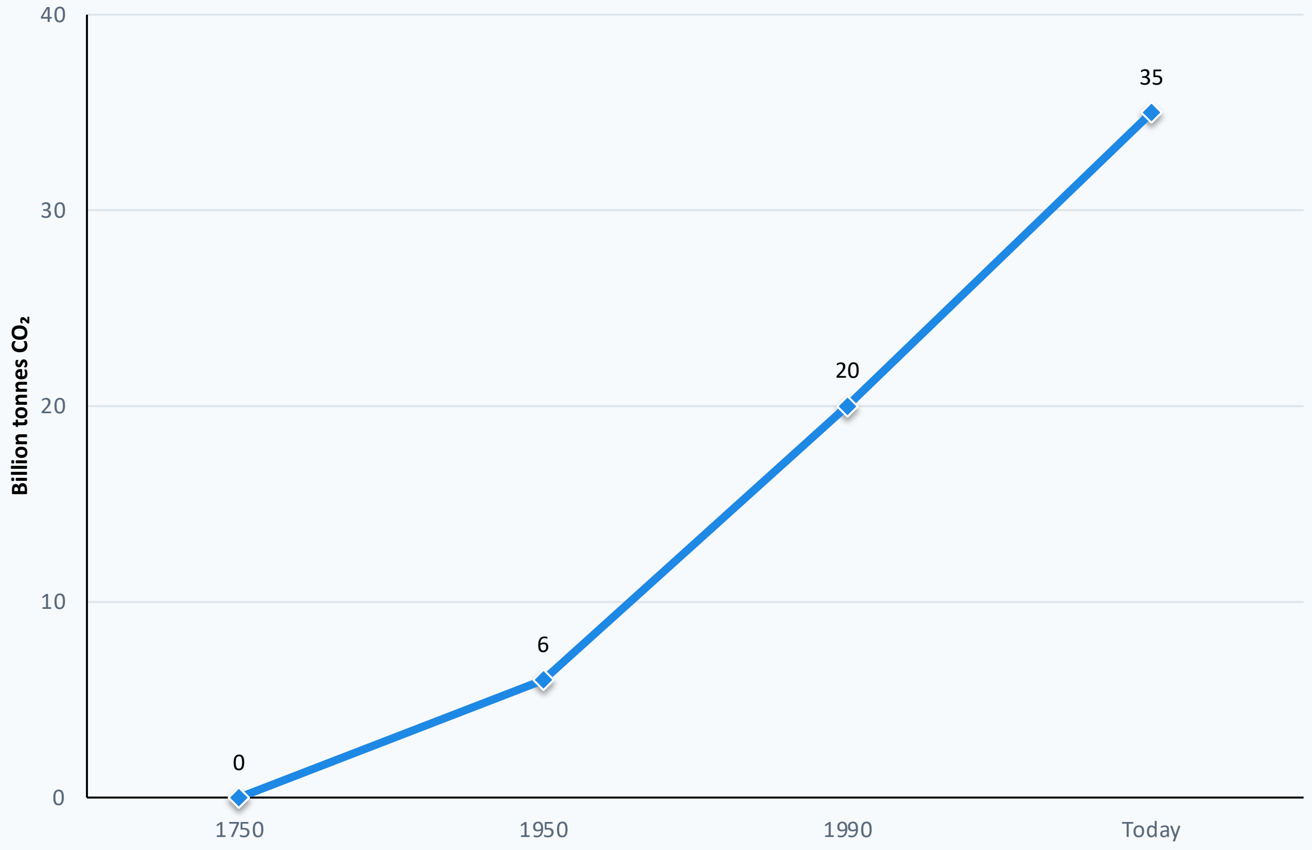
Feb 2026

429 ppm

- Ground-based and satellite measurements confirm the sharp acceleration.
- Primary driver cited in the source brief: burning coal, oil, and natural gas.
- Current level is at least 1.5× greater than the natural increase after the last ice age.

Sources: NASA climate vital signs; Our World in Data / Global Carbon Budget 2025

Global Fossil Fuel CO_2 Emissions Grew Six-Fold: 6 Billion to 35+ Billion Tonnes



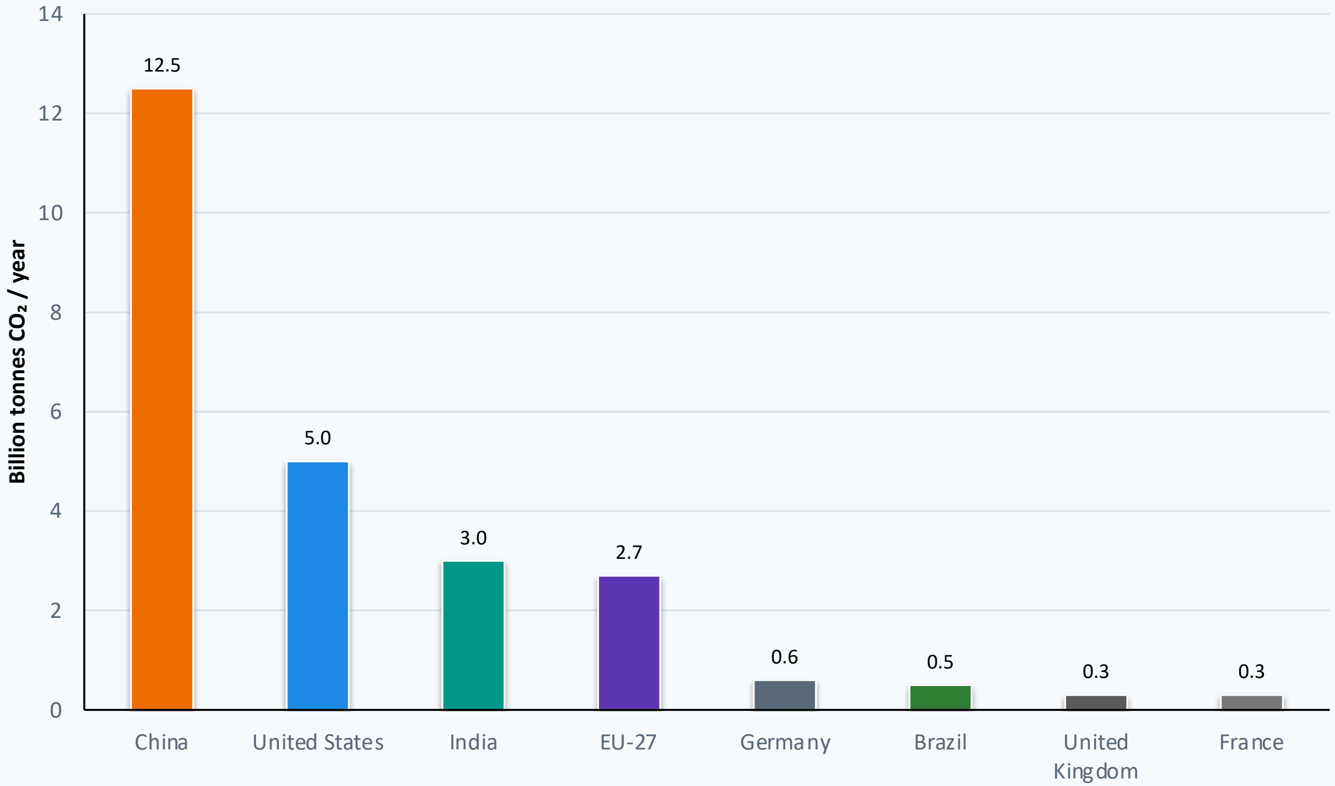
Line chart uses only milestone values reported in the source brief; “Today” denotes current annual fossil fuel and industry emissions exceeding 35 Bt.

Interpretation for negotiators

- Before the Industrial Revolution, emissions were negligible.
- Growth remained slow until the mid-20th century.
- By 1990, global annual emissions had nearly quadrupled from 1950.
- Growth has slowed in recent years, but emissions have not yet peaked.

Reference: figure_2 requested in outline; no figure_2 asset was provided, so the slide uses a native m

China Now Emits Over 12 Billion Tonnes — More Than Double the United States



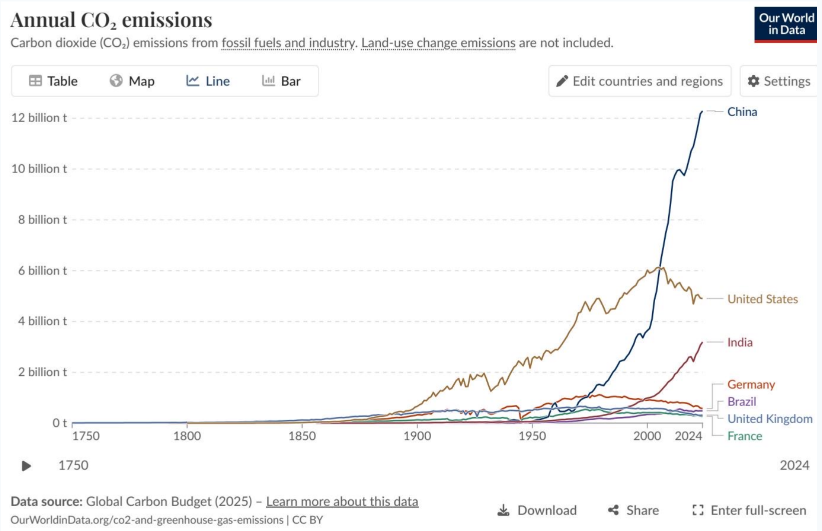


Figure 3: annual CO_2 emissions by country

China: ~12.5 Bt / ~27%; slight increase
United States: ~5.0 Bt / ~14%; declining from ~6 Bt peak around 2005
India: ~3.0 Bt / ~8%; rising

Sources: NASA climate vital signs; Our World in Data / Global Carbon Budget 2025

CO_2 emissions: fossil fuels and industry unless stated

REGIONAL SHARE SHIFT

Asia Now Accounts for ~50% of Global Emissions; US + Europe Below One-Third

Share of global emissions: from industrial concentration to wider geographic distribution

1900

Europe + US >90% of global emissions

- Europe + US: >90%
- Rest of world: <10%

2024

Asia ~50%; US + Europe < one-third

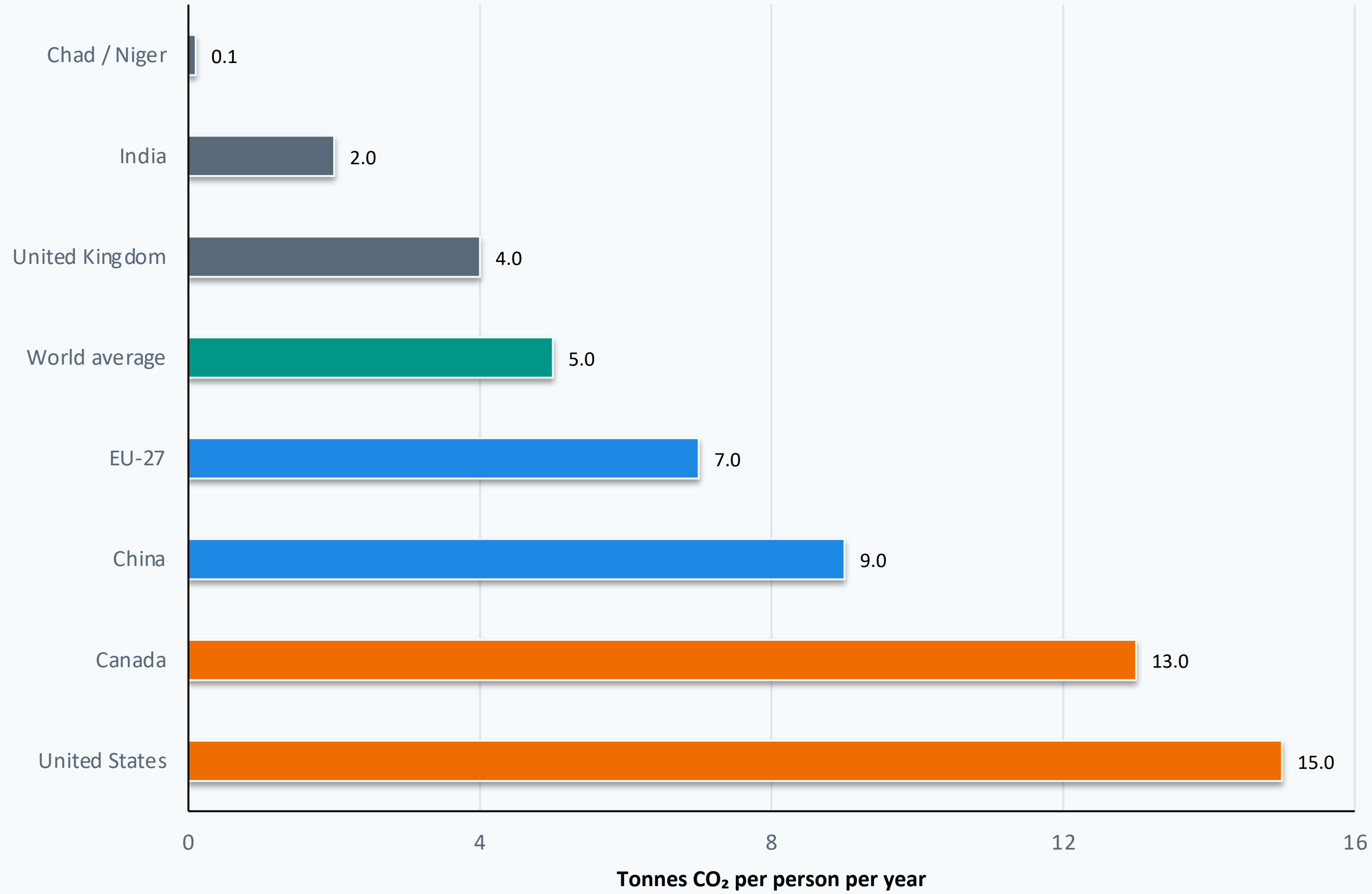
- Asia: 50%
- US + Europe: <33%
- Africa: 3–4% each
- South America: 3–4% each
- Other

By 1950, Europe and the US still represented over 85% of global emissions. Today, Asia produces roughly half while hosting nearly 60% of the world's population.

Sources: NASA climate vital signs; Our World in Data / Global Carbon Budget 2025

CO_2 emissions: fossil fuels and industry unless stated

Per Capita Gap: 15 t in the US vs. 0.1 t in Chad — a 150× Disparity



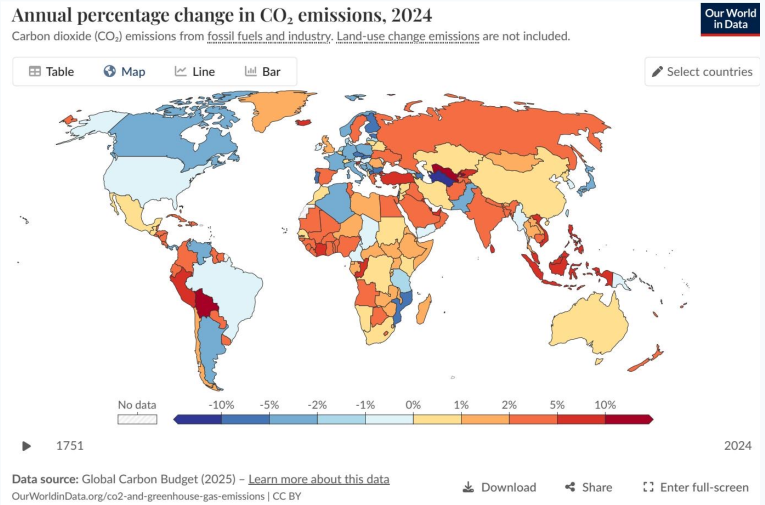


Figure 1: per capita CO_2 emissions over time

Negotiation implication

The average American or Australian emits in under two days wha

Kenya is reported as <1 t/person/year in the source brief.

Sources: NASA climate vital signs; Our World in Data / Global Carbon Budget 2025

CO_2 emissions: fossil fuels and industry unless stated

2024 Snapshot: US and Europe Cut Emissions While Russia and SE Asia Increase

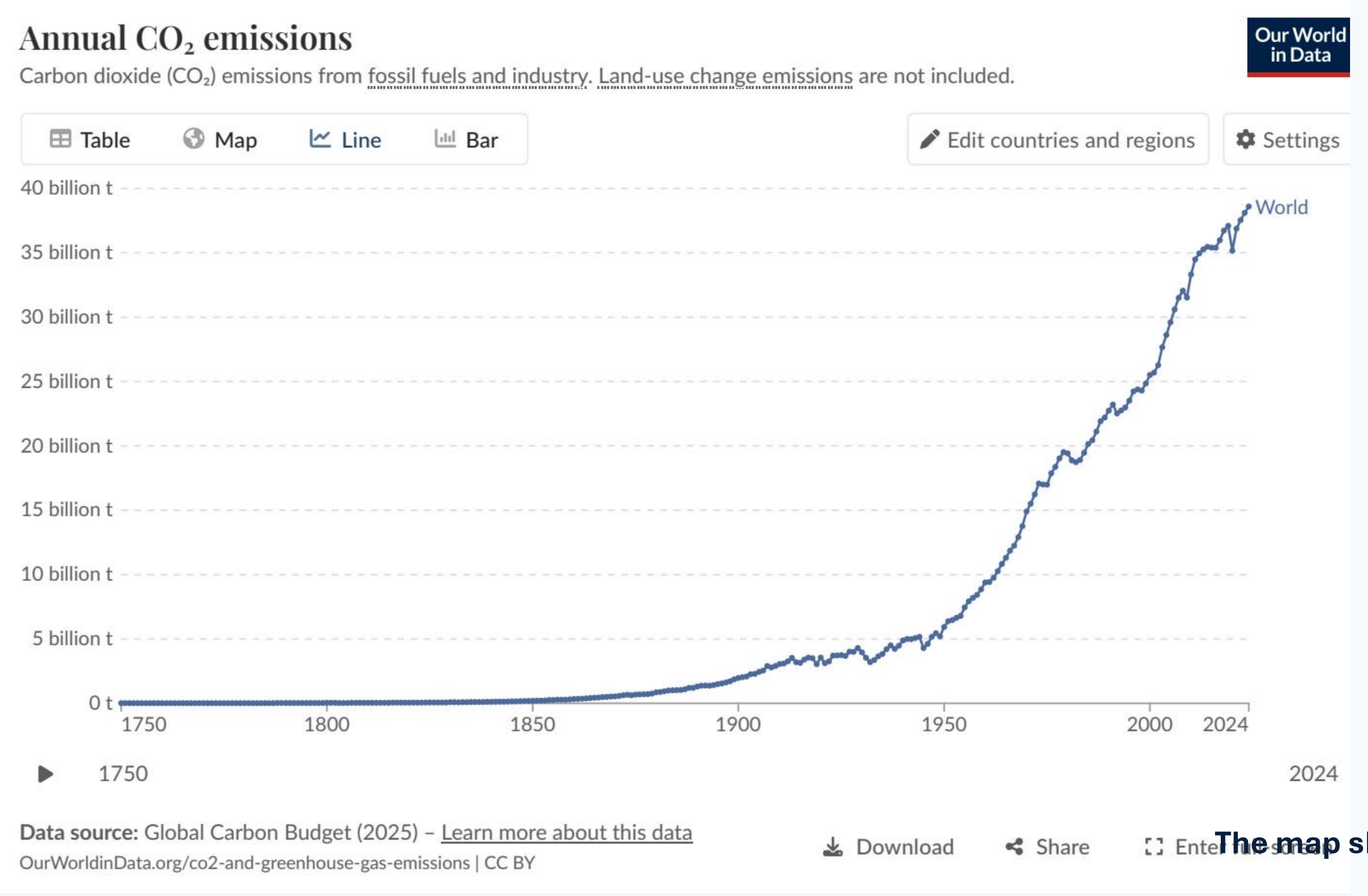


Figure 4: annual percentage change in CO_2 emissions, 2024

Declines

US and much of Europe: roughly −1% to −5% (blue)

Increases

Russia, parts of Southeast Asia, and several African nations: +2% to +1

>10% outliers

Some South American countries recorded very large percentage increa

The map shows mixed short-term momentum, not a global emissions pe

Energy Mix Explains Why France Emits Far Less Per Capita Than Germany

Countries with similar living standards can have very different per capita emissions depending on reliance on nuclear, renewables, or fossil fuels.

Country group	Energy source pattern	Per capita CO_2	Source claim
France / UK	Higher nuclear and renewable shares	Lower; France ~4, UK ~4 t/pe	Brief: much lower than Gerr
Germany / Netherlands	Higher fossil fuel share	Higher than France / UK	Brief: energy mix drives difference

Lower-carbon supply

Nuclear and renewables lower fossil CO_2 intensity.

Fossil exposure

Higher fossil fuel share raises per capita CO_2.

Policy relevance

Prosperity alone does not determine emissions outcomes.

Per capita values explicitly provided: France ~4; UK ~4. Germany and Netherlands are described directionally in the source brief, without numeric per capita values.

KEY TAKEAWAYS

Emissions Are Still Rising, But the Map of Responsibility Is Shifting

35+ Bt

Global annual CO_2 emissions exceed 35 billion tonnes and have not

>25%

China alone accounts for over one-quarter of global emissions (~12.

Regional shift

Europe + US moved from >90% in 1900 to below one-third today. Asia

150×

Per capita inequalities remain extreme: ~15 t/person in the US vs. ~0.

Energy mix

Policy and energy choices shape per capita outcomes, not prosperity alone.

Negotiating lens: total emissions define near-term mitigation scale; per capita emissions and regional shifts define equity and responsibility debates.

Sources: NASA climate vital signs; Our World in Data / Global Carbon Budget 2025